

W13L02

9-May-2024



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How to prove problems NP-complete.

SATISFIABILITY. $f = x_1 \vee x_2 \wedge x_3 \vee \bar{x}_4$

3CNF formula.

↑

3 Conjunctive Normal Form.

a formula is in 3CNF

$$C_1 \wedge C_2 \wedge C_3 \wedge \dots \wedge C_k$$

where each clause has three parts.

$$C_1 = (x_1 \vee x_2 \vee \bar{x}_4)$$

$$C_2 = (x_3 \vee x_4 \vee x_5)$$

$$C_3 = (\bar{x}_2 \vee \bar{x}_3 \vee x_4)$$

$$\phi = (x_1 \vee x_2 \vee \bar{x}_4) \wedge (x_3 \vee x_4 \vee x_5) \wedge (\bar{x}_2 \vee \bar{x}_3 \vee x_4) \\ \wedge (x_2 \vee \bar{x}_3 \vee \bar{x}_5) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_4)$$

SATISFIABILITY for general formulae.

3SAT

for formulae in 3CNF

Clique

$\text{CLIQUE} = \{ \langle G, k \rangle \mid G \text{ is an undirected graph with } k\text{-clique} \}$

we have to show that $\text{CLIQUE} \in \text{NPC}$

Proof: (1) $\text{CLIQUE} \in \text{NP}$

(2) $\forall \pi \in \text{NP} \quad \underline{\pi} \leq_p \underline{\text{CLIQUE}}$

$3\text{SAT} \in \text{NPC}$ ~~therefor~~

$\forall \pi \in \text{NP}, \quad \pi \leq_p 3\text{SAT}$

$$3\text{SAT} \leq_p \text{CLIQUE}$$

Reduction is a transitive relation.

→ $A \leq_p B$ and $B \leq_p C$ then

$$A \leq_p C$$

$\forall \Pi \in NP, \Pi \leq_p 3SAT$ ✓

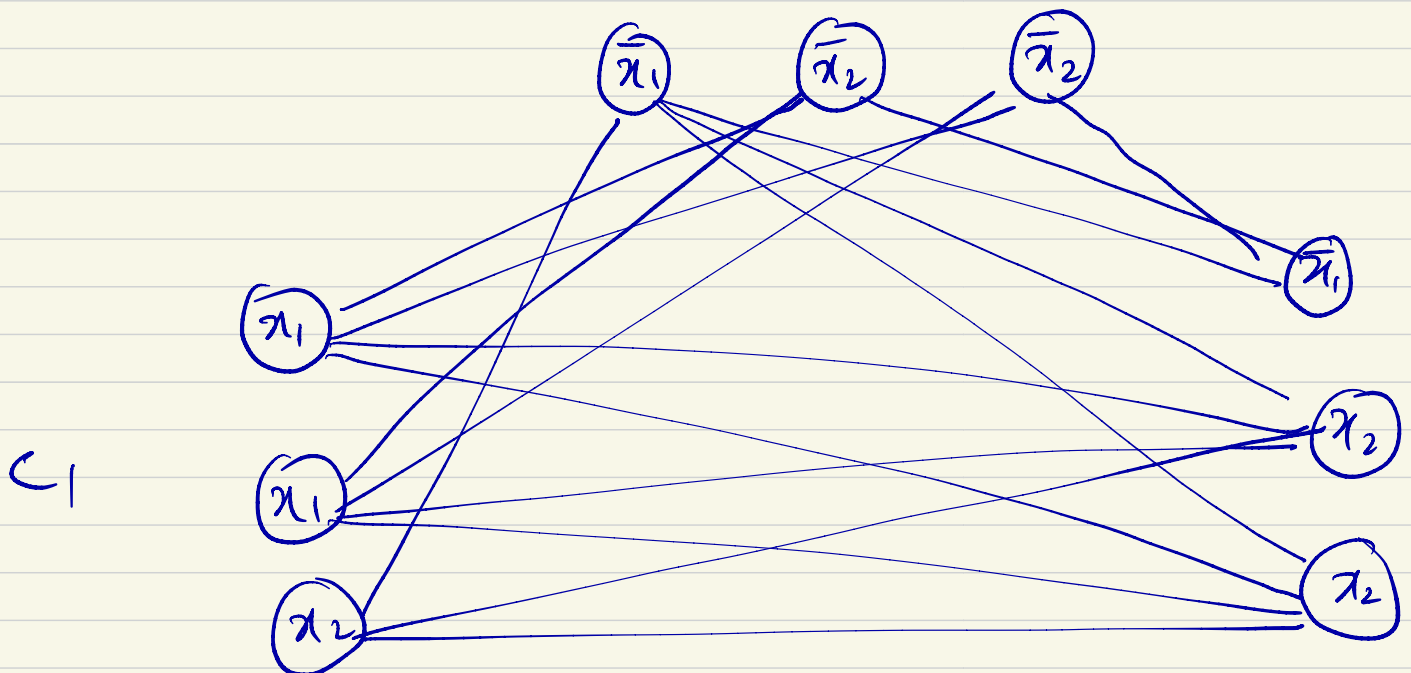
$$3SAT \leq_p CLIQUE$$

⇒ $\forall \Pi \in NP, \Pi \leq_p CLIQUE$.

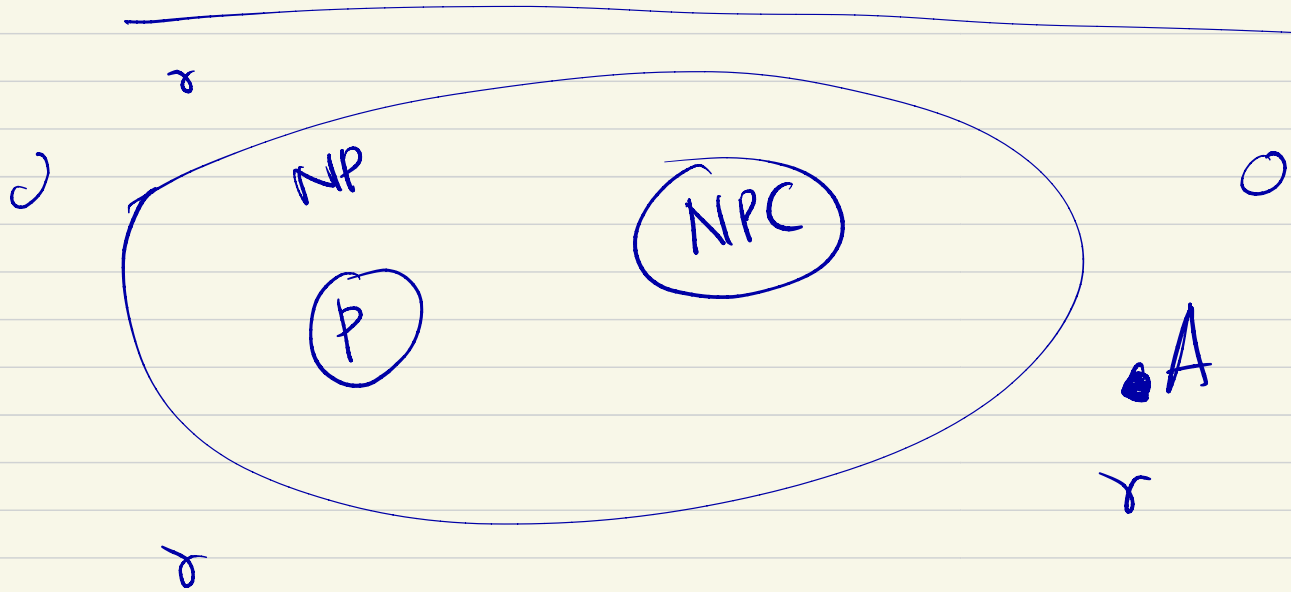
$$3SAT \leq_p CLIQUE$$

$$\phi = (\underbrace{x_1 \vee \bar{x}_1 \vee x_2}_{C_1}) \wedge (\underbrace{\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_2}_{C_2}) \wedge (\underbrace{\bar{x}_1 \vee x_2 \vee x_2}_{C_3})$$

graph



if A is NP-complete and
 we show that A can be
 solved in poly. time then $P=NP$



$\boxed{\text{NP-hard}}$

if $\forall \pi \in \text{NP}$

$\pi \leq_p A$

then A is NP-hard.

NPC: A is NPC iff

① $A \in \text{NP}$

② $\forall \pi \in \text{NP}, \pi \leq_p A$.

NPC: A is NPC iff

① $A \in \text{NP}$. ② A is NP-hard